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Tunnel volume loss study on various geological conditions in Singapore

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ABSTRACT: In a land-scarce country like Singapore, it is inevitable that demands for new space has led to the creation of complex underground networks. As our underground network becomes increasingly built up, the relationship between underground infrastructures in the urban areas also becomes progressively intricate, which presents multifaceted challenges in underground tunnel construction where potential occurrence of ground settlement is the main drawback. With the volume loss induced by tunnel excavation as the main determinant of ground settlement, this paper presents the studies on the volume loss caused by bored tunnelling in different geological conditions in Singapore with a variation of tunnel boring machines (TBM) used in some of the recent Metro line projects such as Downtown Line (DTL), Thomson-East Coast Line (TEL) and Circle Line 6 (CCL6). Back analysis on trough width parameter from the available data is also discussed. The main findings of this study also provide references for impact assessment of bored tunnelling in future projects.

1 INTRODUCTION

The Downtown Line (DTL) and Thomson-East Coast Line (TEL) are the fifth and sixth underground Mass Rapid Transit (MRT) lines which form a boundless journey, bringing people from different parts of Singapore to the central business district. The DTL and TEL consist of 34 and 32 underground stations stretching 41.9 kilometers and 43 kilometers across the island, respectively. The Circle Line 6 (CCL6) closes the Circle Line loop by connecting Marina Bay to HarbourFront stations which the tunnelling has been completed in early 2022. DTL has been in full operation, TEL is currently opening in stages while the recent CCL6 is targeted to open in 2026. With such a comprehensive tunnel construction across the island, this paper studies the settlement induced by tunnelling works of TEL, DTL and CCL6.

The distinctive geological conditions including Bukit Timah Granite (BTG), Jurong Formation (JF), Old Alluvium (OA), Kallang Formation encountered along TEL, DTL and CCL6 alignments as shown in Figure 1 resulted in the use of different types of tunnel boring machine (TBM) based on the respective ground characteristics. The bored tunnelling of TEL, DTL and CCL6 has come across with the four major formations in Singapore where the two main TBMs used are slurry shield and earth pressure balance shield (EPB). Generally, slurry TBMs were used in Bukit Timah Granite Formation to tackle the hard and abrasive layers whereas EPB TBMs were adopted on predominantly cohesive soils.

2 MAJOR GEOLOGY FORMATIONS IN SINGAPORE

2.1 *Granite formation (BTG)*

The Bukit Timah Granite is found at the center of the Singapore Island and at Pulau Ubin (DSTA 2009). The rocks in this formation include acid rocks such as granite, adamellite and granodiorite, and also hybrids of acid and intermediate rocks (granodioritic and dioritic). This formation is intruded by dykes.

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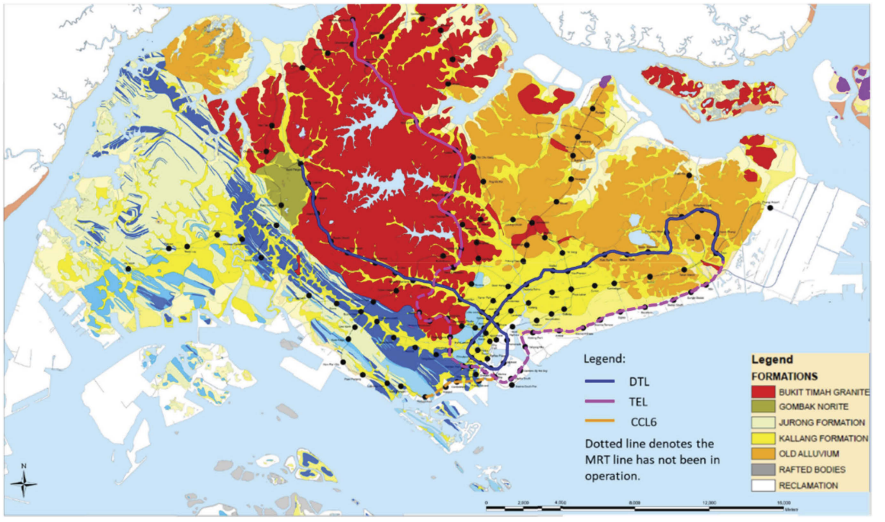


Figure 1. Geology of Downtown Line (DTL), Thomson-East Coast Line (TEL) and Circle Line 6 (CCL6).

2.2 *Jurong Formation (JF)*

The Jurong formation is exposed mostly at the western and southern parts of Singapore. The formation consists of a variety of sedimentary rocks that have been subjected to a variable degree of metamorphism including Sandstones, Siltstones, mudstones, conglomerate and limestone. The rocks of the Jurong Formation exhibit a wide range of strength in the Fresh State and have weathered in different ways.

2.3 *Fort Canning Boulder Bed (FCBB)*

It comprises sandstone boulders in a matrix of stiff over-consolidated silty clay. The deposit was found in the central business district, around Raffles Place and City Hall MRT stations.

2.4 *Old Alluvium (OA)*

The Old Alluvium extends from Southern Johor to east of Singapore. The Old Alluvium is an alluvial deposit that has been variably cemented, often to the extent that it has the strength of a very weak or weak rock. It was found to lie at a depth of 150m below mean sea level, and its thickness can reach up to 195m (DSTA, 2009). The formation consists of clayey sand with pebbles. The sand is coarse, angular and mainly quartzo-feldspathic. The pebbles are predominantly quartz and generally angular. However, ryholite, chert and argillite pebbles are also found, but they are sub-rounded or rounded.

2.5 *Kallang Formation (KF)*

The Kallang formation is the most recent formation and was deposited in river courses, coastlines, and some inland low-lying areas. The Kallang formation often infills sharp, often deeply eroded valleys into the older formations. There are five members recognized within the formation namely marine, alluvial, littoral, transitional and reef. The marine member is locally referred as soft marine clay. The base of the member is usually characterized by a peaty clay and sand layer. Thin sand and peat layers may occur within the sequence. The marine member is sometimes separated by an intermediate layer of reddish-brown clay. This intermediate layer divides them into upper and lower marine members.

3 STUDY ON GROUND SURFACE SETTLEMENT PROFILE

Ground surface settlement profile along DTL, TEL and CCL6 has been studied with the use of LTA Geotechnical Database where key parameters such as ground settlement marker, ring number, date of ring built, geological condition at tunnel axis, tunnel axis, existing ground level, type of TBM and its excavation diameter were retrieved. Upon analysis of the respective settlement profiles, it was observed that immediate settlement occurred within a short period just after the passage of TBM. The immediate settlement generally falls within the acceptable limit of 15mm except under special conditions such as cutterhead intervention, shift change and maintenance works.

4 BACK ANALYSIS

Tunnelling often leads to settlements of the soil surface due to over-excavation, soil relaxation and inefficient tail void filling.

The magnitude of volume loss is influenced by tunnelling management, characteristics of the tunnel boring machines (TBM), and the geotechnical conditions. In predictions of surface settlement (Peck, 1969) and subsurface settlement (Mair et al., 1993), the volume loss is often determined by engineering experience.

4.1 *Study methodology and assumptions*

It is well known that the transverse settlement trough caused by the tunnel excavation may be approximated by a Gaussian distribution curve with the following equation (Peck, 1969):

$$S_v = S_{max} exp\left[-\frac{y^2}{2i^2}\right] \tag{1}$$

where Sv is the vertical settlement of a point lying a horizontal distance y from the tunnel axis, S_{max} is the maximum settlement above the tunnel axis, i is the horizontal distance from the tunnel centreline to the point of inflection of the settlement trough. Figure 3 below shows the typical transverse settlement trough (Gaussian Distribution Curve) caused by tunnelling.

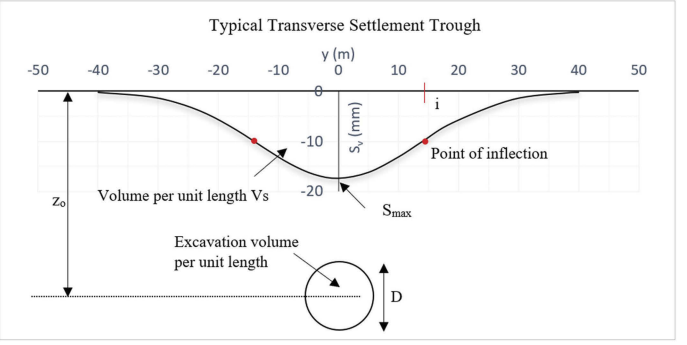


Figure 2. Typical transverse settlement trough by Tunnelling - Gaussian distribution curve (not to scale).

O'Reilly and New (1982) further showed that the trough width i can be estimated as follows:

$$i = K\, z_o \tag{2}$$

where K is the transverse trough width parameter and zo is the depth to the tunnel axis as shown in Figure 3.

4.1.1 *Deriving of Trough Width Parameter K*

The settlement points S and the maximum settlement Smax for each monitoring array can be used to determine the settlement trough width parameter K based on the approach described by Mair et al (1993). Based on Equation 1 & 2, the relationship between $\log_e(S/S_{max})$ and $(x/z_0)^2$ is established as equation below, where x is horizontal distance from the tunnel axis replacing y in Equation 1.

$$\log_e(S/S_{max}) = -0.5/K^2 \cdot (x/z_0)^2 \tag{4}$$

By plotting the points of $(x/z_0)^2$ and $\log_e(S/S_{max})$ as x and y axis, it is able to generate a best fit linear line of the plotted points $[(x/z_0)^2, \log_e(S/S_{max})]$. The value of K can be obtained from the slope of the best fit linear line according to the equation below:

$$K = \sqrt{-0.5/\left[slope\ of\ \log_e(S/S_{max}) versus\ (x/z_0)^2\right]} \tag{5}$$

In Equation 4, the constant C in formula is 0, hence, the best-fit line should start with the point of (0, 0).

4.1.2 *Deriving of Volume Loss*

By integrating Equation 1, the volume of the settlement trough per unit length Vs is obtained by Equation below.

$$Vs = \sqrt{2\pi}\ iS_{max} \tag{6}$$

Where i is obtained from Equation 2 with back-calculated K for various ground conditions and tunnelling methods. Smax is the measured ground settlement above the tunnel axis. Tunnel volume loss is the ratio between the volume of the settlement trough per unit length (Vs) and theoretical TBM excavation volume $(\pi D^2/4)$ per unit length as shown by Equation 7 below:

$$V_L = \frac{4Vs}{\pi D^2}\ x\ 100\% \tag{7}$$

Where D is the excavation diameter of the tunnel (m).
Following assumptions are considered in the back-analysis:

- The "immediate" settlement is only considered for this study.
- Transverse ground settlement profile follows Gaussian Distribution Curve.
- S_{max} considers settle points within 3m of tunnel centre line.
- Generally, 10% of ground settlement points are discarded for K value calculation due to irrelevant readings.

4.2 *Results of study*

10 broad categories were studied based on the ground conditions as listed in Table 1. The category 4 and 6 are further divided into two separated cases with the adoption of different tunnelling method. The tunnelling projects reviewed in each category are also included in Table 1.

The K of best-fit line and the corresponding volume loss of category 1 - Jurong Formation (SIV, SV & SVI) with EPB TBM are plotted as shown in Figure 4. The moderately conservative line of K is indicated in the plots for reference with the K of best-fit line. The numbers of the monitoring array N included in the study for each category are also indicated. The coefficient of determination (R^2) value of the best-fit linear line is reflected to indicate the "goodness of the fit". The average and upper bound values of volume loss are plotted. The points of volume loss are showed in different colors for the source of different projects. The plots for other categories were not included in this paper due to the limitation of maximum number of pages.

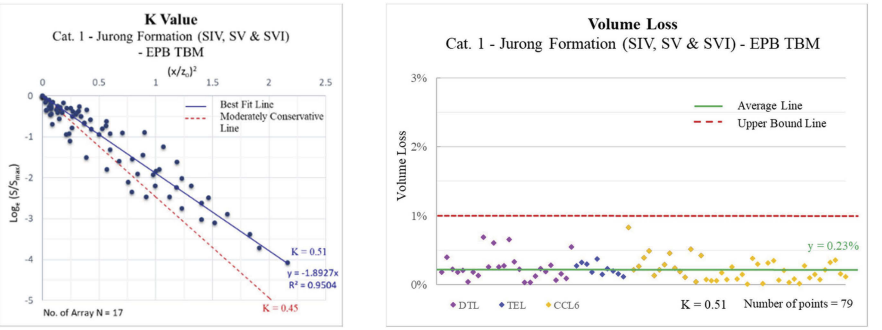


Figure 3. Plots of Trough width parameter (K) and Volume Loss (V_L) for Cat. 1 - Jurong Formation (SIV, SV & SVI) with EPB TBM.

The study results of best-fit line K value, average and upper bound volume loss values for each individual category are summarized in Table 2.

Table 1. Categories of study.

Category	Ground Type	Tunnelling Method	Projects Reviewed
1	Jurong Formation (SIV, SV & SVI)	EPB	DTL2, TEL and CCL6
2	Jurong Formation (SI, SII & SIII)	EPB	DTL2, TEL and CCL6
3	Jurong Formation Mixed Face [a] (Soil/Rock)	EPB	DTL2, TEL and CCL6
4	Bukit Timah Granite (GV & GVI)	Slurry	DTL2 and TEL
		EPB	DTL2 and TEL
5	Bukit Timah Granite (GI, GII & GIII)	Slurry	DTL2 and TEL
6	Bukit Timah Granite Mixed Face [b] (Soil/Rock)	Slurry	DTL2 and TEL
		EPB	DTL2 and TEL
7	Old Alluvium (OA)	EPB	DTL3 and TEL
8	Kallang Formation	EPB	TEL
9	Old Alluvium and Kallang Formation Mixed Face	EPB	DTL3 and TEL
10	Kallang Formation Treated by Grouting	EPB	TEL

a – the mix of any grade between SI, SII & SIII with SIV, SV & SVI.
b – meets any of the following:

- the mix of any grade between GI, GII & GIII with GV & GVI;
- GIV;
- GIV mixes with any grade of GI, GII, GIII, GV & GVI.

4.3 *Trough width parameter*

The trough width parameter K is a constant subjected to soil type and tunnelling method. Generally, sandy soil has smaller K than clayey soil. The K is a critical factor that affects the extent and slope of transverse settlement profile in tunnelling impact assessment. Given constant volume loss, meaning the volume of settlement trough per unit length is fixed, the greater K predicts wider but gentler slope of settlement trough, while the smaller K gives narrower but steeper slope of settlement trough. This is crucial in analyzing the tunnelling impacts to adjacent building and structures which are normally analyzed by differential settlement. As such, the moderately conservative line of K is smaller than the K of best fit line.

In all categories, K value obtained from best-fit line is greater than the K of moderately conservative line (K = 0.45 or 0.5). This can be seen in the plot of $[(x/z_0)^2, \log_e(S/S_{max})]$ that the slope of the best-fit line is gentler than the linear line corresponding to K = 0.45 or 0.5.

The steeper slope of the linear line leads to the smaller K value, which consequently predicts greater rotation of building and structures adjacent to the tunnel. Therefore, a slight smaller K is recommended to deliver moderately conservative approach in the impact assessment of tunnelling effects. In summary, K = 0.45 is hence recommended for most of the ground conditions and K = 0.5 is suggested for soil of Kallang Formation.

4.4 *Volume loss*

With the back-calculated trough width parameter K being obtained for each category, the volume loss can be calculated using Equation 6 & 7 with the known values of measured ground settlement, tunnel depth and tunnel excavation diameter. The results of the average and upper bound volume loss for each category are also summarized in Table 2 below.

There are a few outlier volume loss values which are higher than the upper bound. Common reasons for outlier values were observed as follows:

- TBM stoppage due to TBM cutterhead intervention (CHI) or other maintenance works; and
- Tunnelling in mixed face with presence of thick layer Kallang Formation.

Higher volume loss is suggested at the locations of TBM stoppages, especially with poor ground condition. It is advised that TBM maintenance should be properly planned as such TBM does not stop at the location proximity to critical buildings and utility joints or prior treatment is carried out should TBM have to stop at the critical locations.

4.5 *Characteristic value*

The upper bound of volume loss is commonly adopted in the impact assessment for bored tunnelling works. Considering the scatter of the data, the use of upper bound volume loss may result in very conservative approach in predicting the tunnelling impact, as the gap of interpretation between the upper bound and the actual ground settlement caused by bored tunnelling is large enough to produce significant different design outcomes. The characteristic value of volume loss is hence recommended to reduce the variation of interpretation and provide more representative and reasonable consideration of bored tunnelling effects. This is also in line with Eurocode 7, which defines the characteristic value of a geotechnical parameter as "a cautious estimate of the value affecting the occurrence of the limit state' and suggests that this should be selected with 95% confidence.

4.5.1 *Procedure of determination of characteristic value of volume loss*

To determine of the characteristic value of volume loss with the available data obtained in Section 4.2, the following steps are involved:

1) Calculate the mean value of volume loss $V_{L.m}$ with the available data;
2) Determine the standard deviation σ using equation 9 below.

$$\sigma = \sqrt{\frac{\Sigma_1^n (v_L - v_L)^2}{n-1}} \tag{8}$$

3) Calculate the characteristic value of volume loss $V_{L.ck}$ with a Student t value for 95% confidence level based on the numbers of available data.

$$V_{L.ck} = V_{L.m} + t\sigma \tag{9}$$

Example - determining the characteristic value of volume loss $V_{L.ck}$ for the case of bored tunnelling in Bukit Timah Granite Formation Rock (GI, GII & GIII) with Slurry TBM.

1) the mean value of volume loss $V_{L.m}$ is calculated $V_{L.m}$= 0.30%;

2) Determine the standard deviation σ using equation 9 with all available data, which σ obtains σ = 0.24%;
3) For N=106, student t = 1.64485 for 95% confidence level. Based on equation 10, it is obtained $V_{L.ck}$ = 0.7%

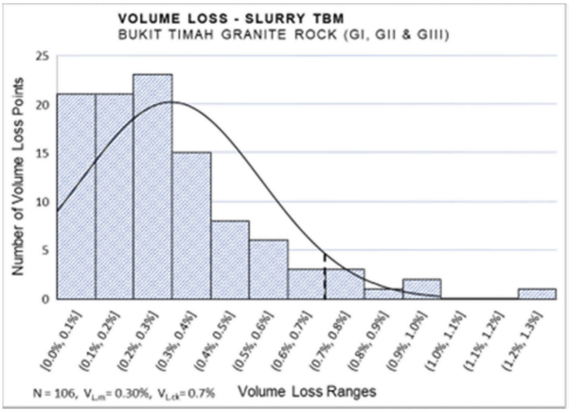


Figure 4. Volume loss distribution for Bukit Timah Granite Formation Rock (GI, GII & GIII) with Slurry TBM.

The 106 volume loss data are grouped in 0.1% intervals (note – the intervals are subjected to the number of data available) and plotted as x axis, and the number of volume loss points fall in each group are plotted as y axis, it is able to obtain the volume loss distribution bar for the case of bored tunnelling in Bukit Timah Granite Formation Rock (GI, GII & GIII) with Slurry TBM as Figure 4 above. The normal distribution of 106 volume loss data is also superimposed in Figure 4. In addition, the normal distribution of the given volume loss data is superimposed into the plot with the same x axis.

4.5.2 *Characteristic value of volume loss*
Similar procedure has applied to all other categories and the characteristic values of volume loss are obtained as summarized in Table 2 below.

It is observed that the volume loss distribution bar for the case of Bukit Timah Granite Formation Rock does not match well with the normal distribution curve. Whereas the volume loss distribution bar and the normal distribution curve are in similar trend for the case of Jurong Formation mixed face. The patterns of volume loss distribution bar are varied with different value of volume loss intervals being adopted. Hence, it may not be consistent with the normal distribution in all cases.

5 SUMMARY AND RECOMMENDATION

Ten broad categories of ground conditions at tunnel face are reviewed in this paper. The results of trough width parameter K, average and upper bound volume loss values and the recommendations for each category are summarized in Table 2 below.

As discussed in section 4.5, tunnel volume loss is provided for the purpose of impact assessment, the characteristic value of volume loss with the K of moderately conservative line is recommended to reduce the variation of interpretation and provide more representative and reasonable consideration of bored tunnelling effects.

However, engineers may adopt the values with the judgement on the concerning factors such as sensitivity of the adjacent building/structures and their confidence on tunnelling performance, e.g. the experience of the tunnel crew with respect to the ground conditions and type/brand of TBM machine and etc.

In addition, higher volume loss should be considered at the locations of TBM stoppages. Else, there is often provided with the mitigation measures at the location of TBM stoppage beforehand to minimize the tunnelling impacts such as to carry out ground treatment at the location of planned CHI.

Table 2. Summary of Study results and recommendations.

S/N	Ground Type	Tunnelling Methods	Result of Study from DTL, TEL & CCL6			Recommendation based on the Study of DTL, TEL & CCL6	
			K (Best fit)	$V_{L.m}$ (%) Average	V_L (%) Upper Bound	K (Moderately Conservative)	$V_{L.ck}$ (%) Characteristic Value*
1	Jurong Formation (SIV, SV & SVI)	EPB TBM	0.51	0.23	1.0	0.45	0.5
2	Jurong Formation (SI, SII & SIII)	EPB TBM	0.50	0.30	1.0	0.45	0.7
3	Jurong Formation Mixed Face (Soil/Rock)	EPB TBM	0.52	0.44	1.5	0.45	1.0
4	Bukit Timah Granite (GV & GVI)	EPB TBM	0.51	0.61	1.5	0.45	1.3
		Slurry TBM	0.49	0.41	1.5	0.45	1.0
5	Bukit Timah Granite (GI, GII & GIII)	Slurry TBM	0.47	0.30	1.0	0.45	0.7
6	Bukit Timah Granite	EPB TBM	0.53	1.32	3.0	0.45	2.7
	Mixed Face (Soil/Rock)	Slurry TBM	0.55	0.71	2.0	0.45	1.7
7	Old Alluvium (OA)	EPB TBM	0.48	0.23	0.8	0.45	0.5
8	Kallang Formation	EPB TBM	0.59	0.55	1.5	0.50	1.2
9	OA mixed with Kallang Formation	EPB TBM	0.60	0.70	2.0	0.50	1.5
10	Kallang Formation Treated by Grouting	EPB TBM	0.58	0.31	1.0	0.50	0.6

* Characteristic value is the 95% confidence level

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